

PROJECT SCOPE DOCUMENT

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1. Executive Summary

This document outlines the scope for the development of a new E-commerce Mobile Marketplace Application. The primary objective is to create a cross-platform mobile application that facilitates transactions between multiple buyers and sellers, supporting a projected user base of 10,000 to 50,000 active monthly users. The project aims for an aggressive delivery timeline of three months, leveraging a modern and scalable technology stack. This initial phase will focus on delivering a Minimum Viable Product (MVP) to establish core marketplace functionality.

2. Core Project Deliverables

The following core components and functionalities will be delivered as part of this project's MVP scope:

Cross-Platform Mobile Applications:

- iOS application compatible with recent iOS versions.

- Android application compatible with recent Android versions.

User Management and Authentication:

- Secure user registration and login for both buyers and sellers.

- User profiles for managing personal information and settings.

- Role-based access control (Buyer, Seller).

Seller Features:

- Product listing management (add, edit, delete products).

- Product catalog display with images, descriptions, pricing, and stock.

- Basic order management for sellers (view pending, confirmed, shipped orders).

Buyer Features:

- Product search and filtering capabilities.
- Product browsing by categories.
- Detailed product pages with descriptions and images.
- Shopping cart functionality (add, remove, update quantities).
- Secure checkout process.
- Order history and status tracking.

Payment Integration:

Integration with a single, common payment gateway (e.g., Stripe, PayPal – to be specified).

Rating and Review System:

- Buyers can rate and review products/sellers.
- Sellers can view ratings and reviews.

Core Backend Services:

- Robust API layer to support mobile applications.
- Database management for users, products, orders, and other marketplace data.
- Basic admin panel for content and user management.

Deployment:

- Application deployment to Apple App Store and Google Play Store.

3. Technical Stack & Architecture Guidelines

The technical architecture will be designed for scalability and maintainability, aligning with the client's preferred technologies:

Mobile Application Development:

Technology: React Native

Approach: Single codebase for both iOS and Android platforms, ensuring consistent user experience and faster development.

Backend Services:

Technology: Node.js

Framework: Express.js (or similar, for API development)

Architecture: RESTful API design, adhering to industry best practices for security and performance.

Hosting: Cloud-based infrastructure (e.g., AWS, GCP, Azure) to support scalability for 10k-50k active users.

Database Management:

Technology: PostgreSQL

Type: Relational Database Management System (RDBMS)

Rationale: Chosen for its reliability, data integrity, and robust support for complex queries, suitable for e-commerce transactions.

General Guidelines:

API-first approach, enabling future integrations and extensibility.

Emphasis on secure coding practices and data protection.

Version control using Git.

Modular and scalable architecture to accommodate future feature enhancements and user growth.

4. Estimated Timeline & Key Milestones

The project is estimated to be completed within a three-month timeframe. This aggressive schedule necessitates a focused MVP approach.

Month 1: Discovery, Design & Core Backend Setup

Week 1-2: Detailed Requirements Gathering & User Story Definition, System Architecture Design, UI/UX Wireframing & Prototyping.

Week 3-4: Database Schema Design, Initial Backend API Development (User Auth, Product Management APIs), Basic Cloud Infrastructure Setup.

Deliverables: Detailed Design Document, UI/UX Prototypes, Core Backend API Endpoints for User & Product Management, Database.

Month 2: Mobile Application Development & Core Features

Week 5-6: React Native App Setup, User Authentication Module (Login, Registration), User Profile Management.

Week 7-8: Product Listing Display (Browse, Search), Seller Product Management (Add, Edit), Shopping Cart Functionality.

Deliverables: Functional Mobile Apps (Buyer & Seller), Core UI Elements, Product Catalog and Cart Features.

Month 3: Advanced Features, Integration, Testing & Deployment

Week 9-10: Checkout Process, Payment Gateway Integration, Order Management (View Orders, Update Status), Ratings & Reviews System.

Week 11: Comprehensive Testing (Unit, Integration, Performance, Security), Bug Fixing, User Acceptance Testing (UAT).

Week 12: Final Review, App Store & Play Store Submission, Initial Deployment.

Deliverables: Fully Integrated MVP Marketplace Application, Completed Testing Reports, App Store Listings.

This timeline is an estimation and will be refined during the detailed planning phase. Any significant scope changes or unexpected complexities may impact the delivery schedule.